

THE AGE OF STEAM AND BIG GUN

THE BATTLE OF TSUSHIMA

In the second half of the 19th century, new technology transformed the world's navies. Wooden sailing ships were replaced by steel warships with coal-fired engines, armed with powerful rifled guns in rotating turrets. The Russians and Japanese fought the first fleet encounter between these formidable vessels in the Tsushima Strait in May 1905.

Russia and Japan went to war in 1904 over their rival ambitions to control Manchuria and Korea. From the outset the Japanese Imperial Navy outclassed the Russian Pacific Fleet based at Port Arthur and Vladivostok. In a bold bid to redress the balance, Russia decided to send a large part of its Baltic Fleet from European waters to East Asia, a gruelling voyage that took seven months. By the time the Russian ships reached the Pacific, Port Arthur had fallen to the Japanese. The Russians had to head for Vladivostok, further north, which meant steaming past Japan. Short of coal, Russian Admiral Zinovy Petrovich Rozhdestvensky chose the shortest route, through the Tsushima Strait between Japan and Korea.

Japanese commander Admiral Togo Heihachiro was on the lookout for the Russian fleet, the progress of which around the world had been a public event. Rozhdestvensky hoped to dash through the Strait under cover of darkness, and he might have succeeded but for a new invention: wireless telegraphy (radio). When a Japanese vessel patrolling the Strait spotted the Russian fleet, it was able to inform Admiral Togo instantly. Fast-moving Japanese cruisers kept in visual contact with the Russians, radioing their position to Togo so he could direct his main force in pursuit. The Japanese fleet had no difficulty intercepting the Russians, as it enjoyed a substantial speed advantage, steaming at around 15 knots against the 6 knots of the Russian vessels. Battle was joined on the afternoon of 27 May.

Both fleets steamed in "line astern" formation – one vessel following another – allowing their turret guns

a maximum field of fire without the risk of hitting a friendly ship. Togo turned his fleet to sail parallel to the Russians, with his flagship *Mikasa* in the vanguard. The exchange of fire began at a range of around 5,500 metres (6,000 yards). Crucially, the Japanese had superior range-finding technology, vital if gunners were to hit a target at such distance. They also had shells more suitable for the conditions: whereas the Russians fired armour-piercing rounds, the Japanese shells were fused to explode on contact. Packed with high explosives, they devastated the superstructure of ships they hit, starting fires and raking the decks with deadly steel splinters. Moreover, the Japanese fleet's speed advantage enabled them to "cross the T", sailing their ships across the front of the Russian line. This manoeuvre brought all their guns to bear on the Russians, while the Russians could only reply with some of their forward guns.

ENDGAME

By nightfall the outclassed Russian fleet had suffered devastation; Admiral Rozhdestvensky, a piece of steel embedded in his skull, was among thousands of casualties. When one of the Russian battleships, the *Borodino*, exploded, a shell striking one of its magazines, 784 of its 785 crew were killed. After dark, the Japanese unleashed their destroyers and torpedo boats upon the fleeing, disorganized enemy. Repeated torpedo runs completed the rout begun by the heavy guns. In total, 17 Russian warships out of the original fleet of 27 were destroyed, including seven battleships. Only three reached Vladivostok.

